

Vaughn Zaayer

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Inquisitive Reed College mathematics graduate with broad experience in both theoretical and applied math and computer science. Leverages strong analytical reasoning to develop effective practical software solutions for real-world use. Keen interest in computational geometry, rendering, and graphics technologies.

Education

Reed College, Bachelor of Arts in Mathematics, Minor in Computer Science Aug. 2021 – May 2026

Relevant Coursework: Real Analysis, Algorithms and Data Structures, Abstract Algebra, Computer Systems, Software Engineering, Cryptography, Linear Algebra, Vector Calculus, Private and Fair Data Analysis, Deep Learning, Parallelism and Concurrency.

Relevant Experience

Reed College Senior Thesis Sep. 2025 – May 2026

Algorithms for Mesh Simplification Using Discrete Differential Geometry Portland, OR

- Conducting year-long independent research in computational geometry, implementing mesh simplification algorithms, formalizing complex properties in discrete geometry, and translating abstract mathematical concepts into practical computational applications.
- Presented and defended a 40-page thesis to faculty demonstrating proofs, implementation details, and outcomes.

Computer Science & Technology Association at Reed (CSTAR) Aug. 2021 – Jan. 2025

Vice President Portland, OR

- Led a team in planning and executing workshops, networking events, and professional development initiatives to promote skill-sharing and community engagement in the Computer Science Department.
- Strengthened student-faculty relationships and mentorship opportunities, enhancing collaboration and departmental outreach.

Lindamood-Bell Learning Processes May 2024 – Aug. 2024

Education Clinician Walnut Creek, CA

- Delivered individualized instruction and developed tailored learning strategies, improving reading and writing outcomes for students with diverse needs.
- Authored detailed progress reports contributing to Stanford research on alternative learning methods and neurological development, supporting academic growth and confidence.

Agile Displays May 2023 – Aug. 2023

Quality Assurance Consultant San Jose, CA

- Developed and executed comprehensive testing plans, producing reproducible bug reports and ensuring software and hardware reliability.
- Collaborated with development teams to track and resolve complex bugs, optimizing products before release.

Introduction to Computer Science (CSCI121), Professor Jim Fix Aug. 2022 – Dec. 2022

Teacher's Assistant Portland, OR

- Supported students in mastering programming fundamentals, algorithms, and data structures through individualized guidance and group instruction, providing constructive feedback on assignments and exams.
- Collaborated with the professor on lesson planning, syllabus design, and project development to improve course structure and student engagement.

Projects

Curvature Computation on the GPU – CUDA, HIP, C++ 2026 | [↗](#)

- Computes the Gaussian curvature of each vertex on a mesh in parallel using a GPU
- Can compile to NVIDIA and AMD GPUs

Mesh Classification using Intrinsically Coarsened Meshes – PyTorch, Computational Geometry 2026 | [↗](#)

- Uses the Intrinsic Curvature Error coarsening algorithm from Liu et al. to coarsen a mesh dataset
- Trains on the vertex curvature values of each mesh to classify them

Skills

Languages: C/C++, Python, Rust, Go, CUDA, HIP, Swift, MIPS Assembly, R, MATLAB, Java, JavaScript

Tools: Postman, Vim/NeoVim, JetBrains Suite, Git/GitHub, Linux, LATEX, gdb/lldb, Google Suite, Microsoft-Suite, R Studio